Distinguishing IgG4-Related Ophthalmic Disease From Graves Orbitopathy

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Purpose: The authors aimed to determine key features of IgG4-related ophthalmic disease (IgG4-ROD) and Graves orbitopathy (GO) to aid in diagnosis.

Methods: The authors retrospectively identified ophthalmology patients seen between June 2009 and November 2013 with clinical overlap of GO and IgG4-ROD. Patient findings were reviewed to characterize the 2 conditions.

Results: Among 8 patients (7 male and 1 female), the mean age was 45.8 years. Time between diagnoses of GO and IgG4-ROD ranged from 1 month to 8 years. Imaging showed enlarged extraocular muscles in all patients. Enlarged infraorbital nerves were seen in 4 patients. Tissue biopsy showed CD20+ lymphocytes with a large proportion of IgG4 plasma cells in 7 of 8 orbital specimens. Six patients had a ratio of IgG4:IgG cells >40%.

Discussion: No pathognomonic clinical findings for GO or IgG4-ROD have been reported, but some key features can help distinguish the conditions. GO is likely if findings include increased thyrotropin receptor antibodies, lid retraction/lid lag, and enlarged extraocular muscles with typical tendon-sparing morphology. Findings suggestive of IgG4-ROD include history of asthma and progressive orbital disease in patients with previous diagnosis of GO, disproportionately large lateral rectus muscle, and enlarged infraorbital nerves. Increased serum IgG4 level and biopsy showing >10 IgG4+ plasma cells/high-power field and IgG4:IgG ratio >40% will support the diagnosis of IgG4-ROD.

Conclusions: GO and IgG4-ROD are complicated inflammatory processes affecting the orbit and present diagnostic challenges. The authors recommend biopsy for patients who do not follow the usual clinical course of GO or have clinical characteristics of IgG4-ROD.

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IgG4-related ophthalmic disease (IgG4-ROD) is a fibroinflammatory condition characterized by a dense lymphoplasmacytic infiltrate with a preponderance of IgG4+ plasma cells and fibrosis.[1] This association between IgG4 and fibroinflammatory disease was first observed in the pancreas, but its association with IgG4 and other organ involvement was only established as a syndrome in 2003.[2] Ophthalmic involvement, primarily in the form of dacryoadenitis, was initially reported in 2007.[3,4] Subsequently, reports have described involvement of the lacrimal gland, orbital soft tissue, extraocular muscles (EOMs), and orbital peripheral nerves.[5] Primary involvement of the EOMs in IgG4-ROD, as well as lacrimal gland enlargement, can lead to diagnostic confusion with Graves orbitopathy (GO), the most common inflammatory disease of the orbit.

GO is the most frequent extrathyroidal manifestation of Graves disease. The diagnosis of GO is clinical because no pathognomonic clinical or laboratory findings have been recognized for the diagnosis. Increased levels of thyrotropin receptor antibodies, eyelid retraction/lid lag, and enlarged EOMs lead to a presumptive diagnosis of GO.[6] In some instances, however, not all features are present for suspected GO, and in others, suspected IgG4-ROD causes enlargement of the EOMs or an enlarged lacrimal gland; thus, diagnostic confusion can result.[7]

The purpose of this study was to describe findings among patients with clinical overlap of IgG4-ROD and GO. The authors aimed to illustrate the diagnostic challenges with these entities and determine key features that may aid in achieving the correct diagnosis.

METHODS

Mayo Clinic Institutional Review Board approval was obtained for this Health Insurance Portability and Accountability Act–compliant research protocol. Because it was a retrospective, observational study, informed patient consent was waived.

The authors searched their electronic database for the records of patients who had diagnoses of GO and IgG4-ROD at some point in their care and were seen in their ophthalmology department at Mayo Clinic, Rochester, Minnesota, between June 2009 and November 2013. For all patients identified, the authors reviewed the health record to characterize symptoms, thyroid status, and radiologic, serologic, and histologic findings that led to the diagnosis of IgG4-ROD. All patients were seen by 1 of 2 ophthalmologists with oculoplastic expertise (J.A.G. and E.A.B.). Clinical data collected included patient age, sex, race, smoking status, medical history, and thyroid-related history, as well as examination findings including vision, pupils, motility, proptosis, lid retraction/lag, and other signs or symptoms of orbital inflammation. The authors collected laboratory values including thyrotropin receptor antibodies, total serum IgG, and serum IgG4. For each patient, if multiple sets of laboratory values were available, the authors chose the earliest test performed or the values correlating with active clinical symptoms.

All but 1 orbital biopsy was performed at Mayo Clinic. Pathologic findings from all orbital biopsies were reviewed at the

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authors' institution by 1 pathologist with expertise in ophthalmic pathologic study (D.R.S.). Biopsies were obtained of the lacrimal gland, EOMs, orbital soft tissue infiltrate, infraorbital nerve, or a combination of these.

The diagnosis of IgG4-ROD was based on the number of IgG4+ plasma cells per high-power field (HPF) and/or the ratio between IgG4+ and IgG+ cells identified by immunohistochemical staining. For each specimen, 3 HPFs with the highest density of IgG4+ plasma cells were selected, and an average number of IgG4+ plasma cells/HPF was calculated. A finding of 10 or more IgG4 cells/HPF was required for the diagnosis of IgG4-ROD, as previously described.[8] As proposed by a consensus conference, a ratio of IgG4+:IgG+ cells >40%, if measured, was also required for diagnosis of IgG4-ROD.[9]

All patients included in this study underwent orbital CT, MRI, or both. The authors characterized enlargement of EOMs, infraorbital nerve, lacrimal gland, and systemic/extraorbital involvement. There were no exclusion criteria.

RESULTS

A total of 8 patients (7 male and 1 female) were identified and included in the study. Five of the patients were part of a larger series previously reported.[5] They were included in the current study because at some point in the course of their disease, they were thought to have GO or Graves disease, or because they had Graves disease and IgG4-ROD concomitantly. Mean age was 45.8 years (range, 28–69 years; Table 1). In 7 cases, GO was originally diagnosed, followed by a presumed diagnosis of IgG4-ROD; mean time between these diagnoses was 3 years (range, 1 month to 8 years). In 1 patient, the diagnoses were made at the same time. Five patients had a medical history significant for a systemic feature of IgG4-related disease, including asthma, sclerosing cholangitis, pancreatitis, lymphadenopathy, and rhinosinusitis. Average follow-up duration was 4.5 years (range, 5 months to 16 years).

The most common presenting symptom was evidence of mass effect (proptosis: range, 11–36 mm) without evidence of inflammation (n = 7) (Table 2). Orbital decompression had been previously performed in 2 patients for presumed GO (patients 1 and 4). Two patients had eyelid retraction and lid lag, and 2 had superior limbic keratopathy. Imaging showed enlarged EOMs in all patients, which was bilateral in 7. The inferior, superior, and medial recti were each involved in 7 patients, and the lateral recti in 6. All patients had sparing of the muscle tendons. Lacrimal gland enlargement was noted in 3 patients (2 bilateral and 1 unilateral). Enlarged infraorbital nerves were seen in 4 patients, all bilaterally. Three patients had well-delineated orbital masses, and 2 had diffuse soft-tissue infiltrate in the orbit.

Tissue biopsy showed CD20+ lymphocytes with a large proportion of IgG4+ plasma cells in 8 of 9 orbital specimens (Table 3). Five patients

TABLE 1. Patient characteristics and laboratory values

Patient/Age, y/Sex	Smoker	Medical history	Thyroid-specific history	Thyrotropin receptor antibodies (RR)	Serum IgG4 (mg/dl) (RR, 2.4–121.0 mg/dl)	Time from diagnosis of GO to IgG4-ROD	Follow-up after diagnosis of IgG4-ROD
Possible past GO without expected clinical course							
1/48/M	No	DM type II; Hypertension	Euthyroid GO	1.9 IU/l (0.0–1.3 IU/l)	36.6	2 y	2 y
2/44/F	Yes	None	Graves disease hyperthyroidism treated with methimazole, then RAI	19% (>16%)	107.0	6 y	8 y
IgG4-ROD initially misdiagnosed as GO							
3/47/M	No	Asthma Sleep apnea Chronic sinusitis	None	Negative	31.8	4 y	3 y
4/43/M	No	Asthma Cervical lymphadenopathy; resolved with corticosteroids	None	Negative	374.0	8 y	4 y
5/69/M	No	Asthma Chronic pancreatitis Sclerosing cholangitis	None	Negative	130.0	4 y	16 y
6/34/M	No	Asthma DM type I Autoimmune polyglandular syndrome Panhypopituitarism Hashimoto thyroiditis Ulcerative colitis	Hashimoto hypothyroidism, replacement therapy	Negative	183.0	1 mo	1 y
Possible concomitant GO and IgG4-ROD							
7/53/M	Yes	Pulmonary sarcoidosis Corticosteroid-induced DM	Diagnosis of Graves hyperthyroidism during work-up for IgG4 level, but no previous history Treated with methimazole and IV corticosteroids	5.1 IU/l (0.0–1.3 IU/l)	78.2	n/a	2 y
8/28/M	No	Asthma Nasal polyps	Hyperthyroid, treated with methimazole	9.21 IU/l (0.0–1.75 IU/l)	347	1 mo	5 mo

DM, diabetes mellitus; F, female; GO, Graves orbitopathy; IgG4-ROD, IgG4-related orbital disease; IV, intravenous; M, male; n/a, not applicable; RAI, radioactive iodine; RR, reference range.

TABLE 2. Physical examination and imaging findings at the time of IgG4-ROD diagnosis

Patient	Proptosis (mm)	Lid retraction	Other eye findings	Bilateral involvement	Orbital mass	Enlarged ION	Enlarged EOMs	Extraorbital mass
Possible past GO without expected clinical course								
1	36/35	Minimal L lid retraction	Bilateral lid lag	Yes	No	R>L	SR, IR, MR	No
2	26/22	No	SLK, trace lag L UL	Yes	Superior intraconal R orbit	No	MR and IR	No
IgG4-ROD initially misdiagnosed as GO								
3	31/36 After rituximab, 18/18	No	None	Yes	No Bilateral soft tissue infiltrate	Bilateral	LR>SR,MR,IR, but all enlarged	No
4	36/32 After rituximab, 19/19	No	SLK, R lid lag	Yes	No	L>R	Bilateral LR>SR>IR> MR>SO>IO	Descending thoracic periaortic lymphadenopathy
5	28/27 After rituximab, 17/16	No	Lagophthalmos	Yes	Multiple, bilateral, intraconal and extraconal	Bilateral	IR>SR>LR>MR but bilateral LR disproportionately large	Multiple paraspinal masses thought to be neurofibromas Skull base and orbital masses extending to the anterior Meckel cave through the cavernous sinuses Involvement of maxillary sinuses, muscles of mastication, and masseter muscles Extension into both pterygopalatine fossae
6	11/11	No	None	Yes	Bilateral lacrimal gland enlargement	No	All enlarged, R MR and R IR most, but bilateral LR disproportionately large	Hilar and mediastinal lymphadenopathy
Possible concomitant GO and IgG4-ROD								
7	27.5/19	Yes	None	No	R infiltrative orbital mass and R lacrimal gland enlargement	No	LR>SR>IR,MR, R SO	No
8	29/22	No	Lagophthalmos, SPK R only, choroidal folds	Yes	Bilateral lacrimal gland enlargement and orbital infiltration	No	Bilateral SR and LR enlargement	Thoracic lymphadenopathy

EOMs, extraocular muscles; GO, Graves orbitopathy; IgG4-ROD, IgG4-related orbital disease; IO, inferior oblique; ION, infraorbital nerve; IR, inferior rectus; L, left; LR, lateral rectus; MR, medial rectus; R, right; SLK, superior limbic keratopathy; SO, superior oblique; SPK, superficial punctate keratopathy; SR, superior rectus; UL, upper lid.

had 11 or more IgG4+ cells/HPF, 2 patients had 10 IgG4+ cells/HPF, and 1 patient had 0 to 5 IgG4+ cells/HPF. Six patients had a ratio of IgG4:IgG cells >40%; in the other 2 patients, IgG cells were not measured.

Treatment included rituximab in all patients (Table 3). Symptoms improved substantially in 6 patients after rituximab treatment, and 2 had progression or minimal improvement. These 2 patients were subsequently treated with intravenous corticosteroids and methotrexate, or mycophenolate mofetil, and had symptom resolution.

Patient Subgroups. Patients were categorized into 3 groups: patients 1 and 2 had a past serologic diagnosis of Graves disease (positive thyrotropin receptor antibodies), but the disease did not follow the expected clinical course for GO and they were later found to also have IgG4-ROD; patients 3, 4, 5, and 6 had IgG4-ROD that was initially missed and incorrectly diagnosed as GO; and patients 7 and 8 had both presumed GO and IgG4-ROD simultaneously (Table 3).

Past Diagnosis of GO. Of the 2 patients with a past diagnosis of GO, 1 was a smoker; neither had a history of asthma. Patient 1 was considered as having "euthyroid GO" for 2 years before the diagnosis of IgG4-ROD, although he had 1 minimally increased value for thyroid-stimulating immunoglobulin. His serum IgG4 level was normal. Clinically, he had minimal lid retraction and lid lag. CT showed enlarged superior rectus (SR), inferior rectus, and medial rectus muscles bilaterally. Because of excess proptosis with a presumptive diagnosis of GO (based on characteristic imaging findings and lid retraction), he underwent orbital decompression. The patient had persistent corticosteroid-dependent orbital congestion for the next several years. Repeat imaging 2 years later showed enlarged infraorbital nerves, which prompted an EOM biopsy that demonstrated features of IgG4-related disease. He had substantial improvement in orbital congestion with rituximab therapy.

Patient 2 had a history of hyperthyroidism treated with methimazole and radioactive iodine 8 years before being seen at the authors'

TABLE 3. Histopathologic findings and treatment

Patient	IgG4+ cells/ HPF	IgG4:IgG	Biopsy/histologic findings	Treatment with oral prednisone	Treatment with rituximab
Possible past GO without expected clinical course					
1	>50	n/a	L superior rectus	Marked decrease in proptosis and diplopia	Improvement; 4 doses
2	30	>40%	R orbital mass Reactive lymphoplasmacytic infiltrate	Some improvement in orbital congestion, but intolerable adverse effects	Improvement; 4 initial doses, then 2 maintenance doses, then stopped
IgG4-ROD initially misdiagnosed as GO					
3	20	n/a	L lacrimal sac Reactive lymphoid hyperplasia	Symptomatic improvement with short courses	Improvement maintenance for 2 y
4	10	>40%	R LR Inflammatory infiltrate with nodular germinal centers	Corticosteroid response/dependence	Complete resolution of symptoms; maintained on every 9-mo dosing
5	100	>40%	L infraorbital nerve Reactive lymphoid hyperplasia with CD20+ infiltrate	No previous treatment with corticosteroids	Initial partial response, with progression in orbits and submandibular gland; more rituximab led to remission
6	5	>40%	R lacrimal gland, R LR, R MR Mild fibrosis and mild lymphoplasmacytic infiltrate	On oral prednisone at baseline for Addison disease, but not a candidate for higher dose because of DM	Eye symptoms improved, but only partial response with renal disease, so began mycophenolate mofetil
Possible concomitant GO and IgG4-ROD					
7	10	>40%	R orbit and lacrimal gland Chronic dacryoadenitis	Could not tolerate because of DM	Progression despite rituximab; some improvement with methotrexate and IV corticosteroids
8	100	>40%	R orbit Lymphoplasmacytic infiltrate	Symptomatic improvement	Recently started on rituximab, with improvement

DM, diabetes mellitus; GO, Graves orbitopathy; HPF, high-power field; IgG4-ROD, IgG4-related orbital disease; IV, intravenous; L, left; LR, lateral rectus; MR, medial rectus; n/a, not available; R, right.

clinic. Serum IgG4 level was within the normal range. On clinical examination, she had asymmetric proptosis, mild orbital congestion, and lagophthalmos with superior limbic keratoconjunctivitis in the right eye (Fig. 1). Her examination findings were believed to be consistent with GO, but the chronology was atypical. CT showed enlarged medial rectus and inferior rectus muscles with serial enlargement of a superior mass involving the SR muscle, which was subsequently biopsied. Both of these patients had atypical clinical courses for GO; therefore, biopsies were performed, which ultimately led to the additional diagnosis of IgG4-ROD.

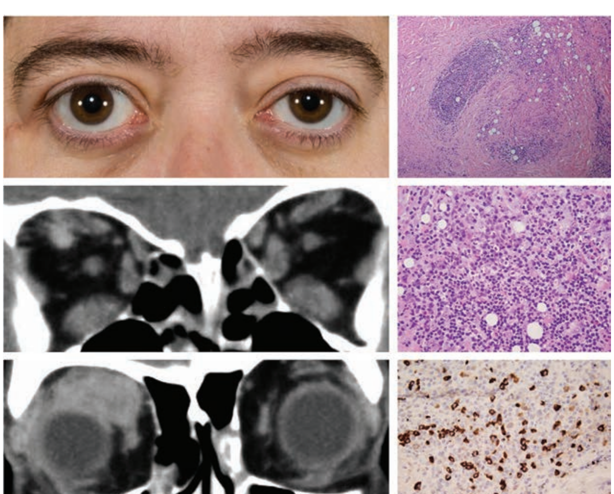

FIG. 1. Patient with possible past Graves orbitopathy that did not follow expected clinical course. Note right-sided proptosis. CT shows enlarged inferior and medial recti with a large mass centered in the right superior rectus. Histopathologic findings of biopsy from right orbital mass show reactive lymphoplasmacytic infiltrate with >30 IgG4+ cells/high-power field. Top, hematoxylin–eosin, original magnification ×40; middle, hematoxylin–eosin, original magnification ×200; bottom, IgG4 immunostain, original magnification ×200.

IgG4-ROD Initially Misdiagnosed as GO. Of the 4 patients with IgG4-ROD that was initially misdiagnosed as GO, all 4 had a history of asthma or other conditions associated with IgG4-related disease, including sclerosing cholangitis, prolonged pancreatitis, pituitary dysfunction, lymphadenopathy, and allergic rhinitis. None were smokers. Three patients had no history of thyroid problems and 1 patient was on replacement therapy for Hashimoto hypothyroidism. All 4 were negative for thyrotropin receptor antibodies. Three patients had increased serum IgG4 levels. On clinical examination, 3 had bilateral exophthalmos greater than 21 mm. No patients had lid retraction; 1 had lid lag (Fig. 2). CT showed bilateral infraorbital nerve enlargement in 3 patients. All patients had disproportionately large lateral rectus (LR) muscles bilaterally.

Time between diagnosis of GO and IgG4-ROD ranged from 1 month to 8 years. Patient 3 had progressive proptosis and orbital congestion that was responsive to corticosteroids. He underwent bilateral decompression at an outside institution for presumed GO. On initial evaluation at the authors' clinic, his pattern of EOM enlargement and progressive clinical course was unusual for GO; therefore, a biopsy was performed. Patient 4 had imaging findings of enlarged EOMs and an expanded orbital fat compartment that was believed to be consistent with euthyroid GO for 8 years. Repeat imaging performed at the authors' institution revealed enlarged infraorbital nerves, which prompted orbital biopsy. Patient 5 had proptosis and imaging findings of enlarged EOMs and infraorbital nerves that predated the discovery of IgG4-related disease. Patient 6 underwent MRI at an outside institution, which was interpreted as EOM enlargement consistent with GO. On evaluation at the

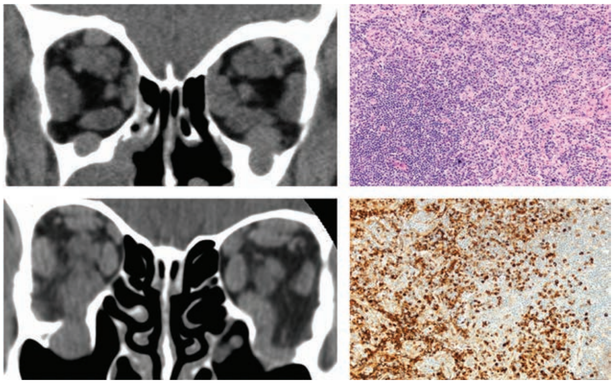

FIG. 2. Patient with IgG4-related ophthalmic disease initially misdiagnosed as Graves orbitopathy. CT shows extraocular muscle enlargement with disproportionately enlarged lateral recti. The infraorbital and frontal nerves are also enlarged. Histopathologic analysis shows reactive lymphoid hyperplasia with CD20+ infiltrate and >100 IgG4+ cells/high-power field. Top, hematoxylin–eosin, original magnification ×100; bottom, IgG immunostain, original magnification ×200.

authors' institution, his clinical examination and imaging findings were atypical for GO, and he underwent orbital biopsy.

Orbital biopsy showed reactive lymphoid hyperplasia and lymphoplasmacytic infiltrate and a ratio of IgG4:IgG cells >40% in all patients. Three patients had 10 or more IgG4+ cells/HPF and 1 (patient 6) had only 5 cells, although this patient was positive for serum IgG4. Three patients had resolution of symptoms with rituximab therapy. Patient 6 had only a partial response to rituximab and was started on mycophenolate mofetil.

Concomitant GO and IgG4-ROD. Patient 7 was found to have increased thyrotropin receptor antibodies during the work-up for IgG4-ROD. He was a smoker and was negative for serum IgG4. His hyperthyroidism was treated with methimazole and intravenous corticosteroids. On clinical examination, he had unilateral proptosis and no lid retraction. CT showed unilateral enlargement of all EOMs with disproportionate enlargement of the LR, and right lacrimal gland enlargement. Because of his lack of lid retraction and LR enlargement, he underwent orbital biopsy. Biopsy of the right lacrimal gland and orbital soft tissue was positive for IgG4, with 10 cells/HPF and IgG4:IgG >40%. He was treated with rituximab but had progression, so treatment was changed to methotrexate, with some improvement.

Patient 8 was a nonsmoker with a history of asthma. He was found to have hyperthyroidism 1 month before diagnosis of IgG4-ROD and was treated with methimazole. He had increased serum IgG4 levels. On examination, he had bilateral proptosis and no lid retraction (Fig. 3). CT showed bilaterally enlarged LR and SR muscles. He underwent orbital biopsy because of atypical clinical and imaging findings. Biopsy of the right orbital soft tissue and lacrimal gland was diagnostic for IgG4; therefore, he was treated initially with oral corticosteroids, which was transitioned to rituximab. He has had marked improvement of his proptosis.

CONCLUSIONS

The 8 patients in this study illustrate the clinical overlap of GO and IgG4-ROD, which can often lead to confusion for the clinician. Although many clinical, laboratory, radiologic, and histologic features support each diagnosis, there are no pathognomonic laboratory or clinical findings. GO remains a clinical diagnosis, and IgG4-ROD may be misdiagnosed as GO. The authors were able to separate these patients into three groups: those with a past possible diagnosis of GO but without the expected clinical course and a later diagnosis of IgG4-ROD, those with IgG4-ROD that was initially missed and incorrectly diagnosed as GO, and those with possible concomitant GO and IgG4-ROD.

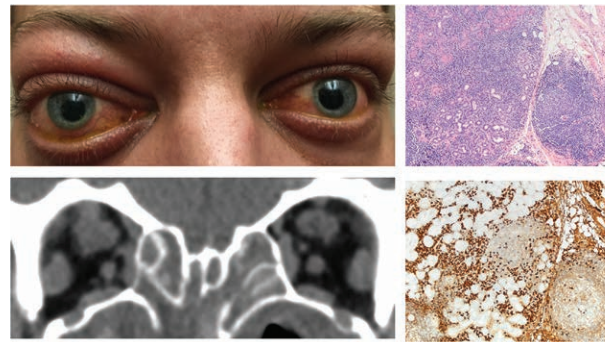

FIG. 3. Patient with possible concomitant Graves orbitopathy and IgG4-related ophthalmic disease diagnoses. Note bilateral proptosis without marked lid retraction. CT shows enlarged superior and lateral rectus muscles and enlarged infraorbital and frontal nerves. Histopathologic analysis of the right orbital biopsy shows lymphoplasmacytic infiltrate with >100 IgG4+ cells/high-power field. Top, hematoxylin–eosin, original magnification ×100; bottom, IgG4 immunostain, original magnification ×100.

GO and IgG4 disease concomitantly affecting the orbit has been previously reported[10,11]; 2 of the patients had presumed concurrent diagnoses of GO and IgG4-ROD on the basis of thyroid laboratory values and a positive biopsy for IgG4. Because GO is a clinical diagnosis, it is possible that some of the patients did not have GO even though they had Graves disease. However, it is also possible that GO and IgG4-ROD may occur concomitantly, because many autoimmune diseases overlap. This is difficult to prove, however, given the absence of a specific diagnostic test for GO. Two patients in the cohort had a previous diagnosis of GO, which was based on positive laboratory values and clinical findings. Two patients in the study had previously undergone orbital decompression for presumed GO and subsequently were found to have IgG4-ROD instead. Four patients in the study had misdiagnoses of GO, all of whom had systemic features of IgG4-related disease. These patients also had findings atypical for GO, including the absence of lid retraction, orbital masses not involving the lacrimal gland, lymphadenopathy, and enlarged infraorbital nerves. The reasons for misdiagnosis may be related to the timing of presentation predating the discovery of IgG4-related disease, as well as dependence on radiographic interpretation that equated EOM enlargement with GO.

Diagnostic criteria have been suggested for IgG4-ROD. Serum diagnostic criteria include serum IgG4 concentration >135 mg/dl.[12,13] Histopathologic diagnostic criteria include a ratio of IgG4/IgG cells of >40% and more than 10 IgG4+ plasma cells/HPF.[9] Two of 3 major histologic features should be present for diagnosis of IgG4-ROD, including a lymphocytic and plasmacytic infiltrate, obliterative phlebitis, and storiform fibrosis, although the latter 2 findings are uncommonly seen in orbital biopsies.[9]

In this study, all patients with biopsy-proven IgG4-ROD had signs of mass effect (proptosis) without evidence of inflammation. This finding aligns with a study by Tiegs-Heiden et al.,[5] which found proptosis or periorbital edema in 96% of patients with IgG4-ROD. Only 1 patient in this study did not have proptosis; interestingly, this patient (patient 6) had negative tissue biopsy findings but had increased serum IgG4 levels, as well as asthma, chronic sinusitis, mediastinal lymphadenopathy, and bilaterally enlarged LRs and lacrimal glands. Patient 6 also was the only patient with a history of hypothyroidism (Hashimoto thyroiditis). As shown in previous studies, patients with IgG4-ROD may be more likely to have bilateral orbital involvement,[8,14] as was seen in 7 patients in this cohort. Bilateral disease, however, also is often seen in patients with GO. Lid retraction, a characteristic finding of GO, was uncommon in patients with IgG4-ROD (2 patients in this study, both of whom had hyperthyroidism and increased thyrotropin receptor antibodies).

In this study, 4 patients had increased serum IgG4 levels (>135 mg/dl). Of these 4, 1 had a history of hyperthyroidism, 1 had hypothyroidism, and 2 had no thyroid-specific history. Patients with Graves disease have been shown to have higher serum IgG4 levels than normal healthy controls. Also, IgG4 levels are significantly higher in patients with GO than in those with Graves disease but no eye involvement,[15] which suggests that some cases diagnosed as Graves disease may belong to the clinical spectrum of IgG4 thyroiditis or IgG4-related disease.[16]

Patterns of imaging findings of IgG4-ROD include LR enlargement out of proportion to the other rectus muscles and LR enlargement in combination with enlargement of any other single muscle, muscle tendon sparing, enlarged lacrimal glands, and enlarged infraorbital nerves. In a study of 27 patients with IgG4-ROD, the LR was the most commonly involved EOM.[5] This can be used as a distinguishing feature of IgG4-ROD because the LR tends to be spared until late in the disease process of GO. Infraorbital nerve involvement in IgG4-ROD also is not typically present in GO.[17] In this study, 5 patients had disproportionately large LRs and 4 had bilaterally enlarged infraorbital nerves, findings unlikely to be associated with GO.

Histologically, GO differs substantially from IgG4-ROD. In GO, glycosaminoglycans accumulate in the body of the EOMs. GO has minimal chronic inflammation, but active GO can have infiltration with primarily CD4+ T cells.[18]

Diagnostic histopathologic features of IgG4-related disease are (1) dense lymphoplasmacytic infiltrate, (2) storiform fibrosis, and (3) obliterative phlebitis.[9] Reactive lymphoid hyperplasia was seen in 7 of the patients, as has been reported previously,[8] but none had evidence of obliterative phlebitis. This is consistent with other reports of the histopathologic characteristics of ocular adnexal IgG4-related disease in which obliterative phlebitis is less common compared with other sites of IgG4-related disease such as the pancreas.[19,20] The appropriate cutoff number of IgG4+ plasma cells to diagnose IgG4-ROD has been controversial. Initially, >10 cells/HPF was thought to be diagnostic,[21] but this has since been subdivided on the basis of organ involvement. In this study, 1 patient had <10 cells/HPF, 2 had 10 cells/HPF, and 5 had >10 cells/HPF. Six patients had biopsies with a ratio of IgG4:IgG cells >40%. A ratio >40% has been suggested as an acceptable cutoff value in any organ.[9,18,22]

There are several limitations of this study. As with any retrospective study, the accuracy and completeness of data rely on the medical record. In addition, the small sample size may not be truly representative: 7 patients in this study were white, so differences in the presentation of GO and IgG4-ROD in other populations may be overlooked.

GO and IgG4-ROD are complicated inflammatory processes affecting the orbit and present diagnostic challenges to the clinician. Although there are no pathognomonic findings, several key features may help the clinician distinguish between the 2 conditions. GO may be suspected with increased thyrotropin receptor antibodies, presence of lid retraction/lid lag, enlarged EOMs with typical tendon-sparing morphology, and combinations of inferior rectus, medial rectus, and SR involvement. Findings suggestive of IgG4-ROD include a history of asthma and progressive orbital disease in patients previously suspected of having and treated for GO.[23] Physicians should complete a careful history, asking for associated features that may occur with IgG4-related disease, as well as a general regional examination for enlargement of lymph nodes and salivary glands in the head and neck. Corticosteroid dependency with noninflamed eyes is not typical of GO and should lead the clinician to consider IgG4-ROD. Radiographic findings suggestive of IgG4-ROD include disproportionately large LRs, especially without involvement of the SR, enlargement of any 2 muscles that includes the LR, and enlarged infraorbital nerves. Histologically, GO does not show fibrosis, and chronic inflammation is minimal compared with IgG4-ROD. Finally, increased serum IgG4 levels, and a typical lymphoplasmacytic fibrosclerotic biopsy with >10 IgG4-staining plasma cells/HPF and IgG4:IgG >40% supports the diagnosis of IgG4-ROD. The authors recommend biopsy for patients whose condition does not follow the usual clinical course or who have clinical characteristics of IgG4-ROD.

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Consent to use clinical photographs was obtained from the patients and is on file.

REFERENCES

1. Stone JH, Khosroshahi A, Deshpande V, et al. Recommendations for the nomenclature of IgG4-related disease and its individual organ system manifestations. *Arthritis Rheum* 2012;64:3061–7.
2. Kamisawa T, Funata N, Hayashi Y, et al. A new clinicopathological entity of IgG4-related autoimmune disease. *J Gastroenterol* 2003;38:982–4.
3. Cheuk W, Yuen HK, Chan JK. Chronic sclerosing dacryoadenitis: part of the spectrum of IgG4-related sclerosing disease? *Am J Surg Pathol* 2007;31:643–5.
4. Takahira M, Kawano M, Zen Y, et al. IgG4-related chronic sclerosing dacryoadenitis. *Arch Ophthalmol* 2007;125:1575–8.
5. Tiegs-Heiden CA, Eckel LJ, Hunt CH, et al. Immunoglobulin G4-related disease of the orbit: imaging features in 27 patients. *AJNR Am J Neuroradiol* 2014;35:1393–7.
6. Bartley GB, Gorman CA. Diagnostic criteria for Graves' ophthalmopathy. *Am J Ophthalmol* 1995;119:792–5.
7. Bartalena L, Chiovato L. Graves'-like orbitopathy: do not forget IgG4-related disease. *J Endocrinol Invest* 2014;37:1233–5.
8. Plaza JA, Garrity JA, Dogan A, et al. Orbital inflammation with IgG4-positive plasma cells: manifestation of IgG4 systemic disease. *Arch Ophthalmol* 2011;129:421–8.
9. Deshpande V, Zen Y, Chan JK, et al. Consensus statement on the pathology of IgG4-related disease. *Mod Pathol* 2012;25:1181–92.
10. Kubota T, Moritani S, Terasaki H. Ocular adnexal IgG4-related lymphoplasmacytic infiltrative disorder and Graves ophthalmopathy. *Arch Ophthalmol* 2011;129:818–9; author reply 819.
11. Bijlsma WR, Kalmann R. Idiopathic orbital inflammation and Graves ophthalmopathy. *Arch Ophthalmol* 2010;128:131–2.
12. Umehara H, Okazaki K, Masaki Y, et al. Comprehensive diagnostic criteria for IgG4-related disease (IgG4-RD), 2011. *Mod Rheumatol*. 2012;22:21–30.
13. Masaki Y, Kurose N, Yamamoto M, et al. Cutoff values of serum IgG4 and histopathological IgG4+ plasma cells for diagnosis of patients with IgG4-related disease. *Int J Rheumatol* 2012;2012:580814.
14. Wallace ZS, Khosroshahi A, Jakobiec FA, et al. IgG4-related systemic disease as a cause of "idiopathic" orbital inflammation, including orbital myositis, and trigeminal nerve involvement. *Surv Ophthalmol* 2012;57:26–33.
15. Bozkirli E, Bakiner OS, Ersozlu Bozkirli ED, et al. Serum immunoglobulin G4 levels are elevated in patients with Graves' ophthalmopathy. *Clin Endocrinol (Oxf)*. 2015;83:962–7.

16. Takeshima K, Inaba H, Furukawa Y, et al. Elevated serum immunoglobulin G4 levels in patients with Graves' disease and their clinical implications. *Thyroid* 2014;24:736–43.
17. Hardy TG, McNab AA, Rose GE. Enlargement of the infraorbital nerve: an important sign associated with orbital reactive lymphoid hyperplasia or immunoglobulin g4-related disease. *Ophthalmology* 2014;121:1297–303.
18. Bahn RS. Graves' ophthalmopathy. *N Engl J Med* 2010;362:726–38.
19. Sato Y, Ohshima K, Ichimura K, et al. Ocular adnexal IgG4-related disease has uniform clinicopathology. *Pathol Int* 2008;58:465–70.
20. Stone JH, Zen Y, Deshpande V. IgG4-related disease. *N Engl J Med* 2012;366:539–51.
21. Chari ST. Diagnosis of autoimmune pancreatitis using its five cardinal features: introducing the Mayo Clinic's HISORt criteria. *J Gastroenterol* 2007;42(suppl 18):39–41.
22. Cheuk W, Chan JK. IgG4-related sclerosing disease: a critical appraisal of an evolving clinicopathologic entity. *Adv Anat Pathol* 2010;17:303–32.
23. Baqir M, Garrity JA, Vassallo R, et al. Asthma and orbital immunoglobulin G4-related disease. *Ann Allergy Asthma Immunol* 2016;116:313–6.